topbook_harder

Khanh Nguyen

June 2023

some harder problems in topbook2023.pdf

Common Theorems

I will put here some common theorems used in this notes.

Theorem 1 (Cover Theorem) *Given a set A, for all element $x \in A$ if $x \in U_x \subseteq A$ then*

$$\bigcup_{x \in A} U_x = A$$

Proof

We immediately have $\bigcup_{x \in A} U_x \subseteq A$. On the other hand, for all $x \in A$, $\{x\} \subseteq U_x$. Then

$$A = \bigcup_{x \in A} \{x\} \subseteq \bigcup_{x \in A} U_x$$

Exercise 1.1.9

Exercise 2.3.4

Let $C[0,1]$ be the set of all continuous real-value functions on $[0,1]$

- Show that the collection $\mathcal{M} = \{M(f,\epsilon) : f \in C[0,1] \wedge \epsilon > 0\}$ where $M(f,\epsilon) = \{g : g \in C[0,1] \wedge \int_0^1 |f-g| < \epsilon\}$ is a basis for a topology $\mathcal{T}_1$ on $C[0,1]$
- Show that the collection $\mathcal{U} = \{U(f,\epsilon) : f \in C[0,1] \wedge \epsilon > 0\}$ where $U(f,\epsilon) = \{g : g \in C[0,1] \wedge \sup_{x \in [0,1]} |f(x) - g(x)| < \epsilon\}$ is a basis for a topology $\mathcal{T}_2$ on $C[0,1]$
- Prove that $\mathcal{T}_1 \neq \mathcal{T}_2$

Let's generalize the first two questions a bit.

Lemma 1 *Let* $\mathcal{B} = \{B(f,\epsilon) : f \in C[0,1] \wedge \epsilon > 0\}$ *where* $B(f,\epsilon) = \{g : g \in C[0,1] \wedge d(f,g) < \epsilon\}$ *such that* d *is a pseudo-metric, i.e: (1) semi-definiteness* $d(a,a) = 0$ *(2) symmetry* $d(a,b) = d(b,a)$ *(3) triangle inequality* $d(a,c) \leq d(a,b) + d(b,c)$.

$\mathcal{B}$ *generates a topology* $\mathcal{T}$ *on* $C[0,1]$.

Remark 1 $f \in B(f,\epsilon)$ *for all* $f \in C[0,1]$ *and* $\epsilon > 0$

Proof of Lemma 1

In order to prove $\mathcal{B}$ generates a topology $\mathcal{T}$ on $C[0,1]$, we need to prove two properties (1) $C[0,1] = \bigcup_{B\in\mathcal{B}} B$ and (2) for any $B_1, B_2 \in \mathcal{B}$, for all $f \in B_1 \cap B_2$, there exists a $B_3 \in \mathcal{B}$ such that $f \in B_3 \subseteq B_1 \cap B_2$

For any $f \in C[0,1]$, $f \in B(f,1) \in \mathcal{B}$. On the other hand, $B(f,\epsilon) \subseteq C[0,1]$ for all $f \in C[0,1]$ and $\epsilon > 0$. Hence, (1)

(2)

Let $B_1 = B(f_1,\epsilon_1)$ and $B_2 = B(f_2,\epsilon_2)$.

For any $f \in B_1 \cap B_2$, we have $d(f_1,f) < \epsilon_1$ and $d(f_2,f) < \epsilon_2$. Hence, choose a positive $\epsilon = \min\{\epsilon_1 - d(f_1,f), \epsilon_2 - d(f_2,f)\}$ and $B_3 = B(f,\epsilon)$

For any $g \in B_3$,

$$\begin{aligned}
d(f_1,g) &\leq d(f_1,f) + d(f,g) && \text{(triangle inequality)}\\
&< d(f_1,f) + \epsilon && (g \in B_3)\\
&\leq d(f_1,f) + (\epsilon_1 - d(f_1,f)) && \text{(choice of } \epsilon)\\
&= \epsilon_1
\end{aligned}$$

Similarly, $d(f_2,g) < \epsilon_2$. Therefore, $f \in B_3 \subseteq B_1 \cap B_2$, So (2)

Main proof

Now, we apply Lemma 1 for $\mathcal{M}$ and $\mathcal{U}$.

For any $x \in [0,1]$ by *Cauchy–Schwarz inequality*, $|a(x) - c(x)| \leq |a(x) - b(x)| + |b(x) - c(x)|$

From the properties of *Riemann integral*,

$$\int_0^1 |a-c| \leq \int_0^1 |a-b| + \int_0^1 |b-c|$$

Furthermore, the LHS is upper-bounded by the RHS, hence their supremums

$$\sup_{x\in[0,1]} |a(x) - c(x)| \leq \sup_{x\in[0,1]} |a(x) - b(x)| + \sup_{x\in[0,1]} |b(x) - c(x)|$$

In the last question, it is obvious that for any $n > 0$ there exists a function $f \in M(0,1)$ such that $\min_{x\in[0,1]} f(x) = -n$ and $\max_{x\in[0,1]} f(x) = n$. If f is in any member $U(f_2,\epsilon_2)$ of $\mathcal{U}$, it must be that $\epsilon > n$. So, there exists a function $g \in U(f_2,\epsilon_2)$ with $\int_0^1 |0-g| = n > 1$ or $U(f_2,\epsilon_2) \setminus M(0,1) \neq \varnothing$. By Proposition 2.3.4 in the book, $\mathcal{T}_1 \neq \mathcal{T}_2$

Exercise 3.1.5.v

Exercise 3.2.9

Let S be a dense subset of a topological space $(X, \mathcal{T})$. Prove that for every open subset U of X, $\overline{S \cap U} = \overline{U}$

Lemma 2 *Let A, B be subsets of a topological space $(X, \mathcal{T})$, then $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$*

Proof of Lemma 2

We will prove (1) $A \cap B \subseteq \overline{A} \cap \overline{B}$ and (2) limit points of $A \cap B$ is in $\overline{A} \cap \overline{B}$

(1)

Any point in A is in $\overline{A}$, any point in B is in $\overline{B}$. Then any point in A and B is in $\overline{A}$ and $\overline{B}$. Hence (1)

(2)

Let $x \in X$ be a limit point of $A \cap B$, so any open set containing x contains a point in $A \cap B$. Hence, any open set containing x contains a point in A, that implies x is a limit point of A, $x \in \overline{A}$. Similarly, $x \in \overline{B}$. Therefore, (2)

Main proof

Apply Lemma 2, $\overline{S \cap U} \subseteq \overline{U}$

Now, we will prove that $\overline{U} \subseteq \overline{S \cap U}$, i.e (1) $u \in U \implies u \in \overline{S \cap U}$ and (2) x is limit point of $U \implies x \in \overline{S \cap U}$

(1)

Let $u \in U$. For any open set O containing u,

S is dense, so u is a limit point of S. Given the open set $U \cap O$ containing u, it must also contains a point $s \in S$. Hence $s \in S \cap (U \cap O) = (S \cap U) \cap O$. Therefore, for any open set O containing u, the intersection of $S \cap U$ and O is non-empty by the construction of s. So, u is a limit point of $S \cap U$

(2)

Let $x \in X$ be a limit point of U. For any open set O containing x, take $u \in U \cap O$,

S is dense, so u is a limit point of S. Given the open set $U \cap O$ containing u, it must also contains a point $s \in S$. Hence $s \in S \cap (U \cap O) = (S \cap U) \cap O$. Therefore, for any open set O containing x, the intersection of $S \cap U$ and O is non-empty by the construction of s. So, x is a limit point of $S \cap U$

Exercise 3.2.11.v

Let $\mathcal{B} = \{[a, b) : a \in \mathbb{R}, b \in \mathbb{Q}, a < b\}$, $\mathcal{B}$ is a basis for a topology $\mathcal{T}_1$ on $\mathbb{R}$, namely the *Sorgenfrey line*. Prove that the *Sorgenfrey line* does not satisfy the second axiom of countability, i.e $(\mathbb{R}, \mathcal{T}_1)$ cannot be generated by a countable number of open sets.

TODO

Proposition 3.3.3

The only clopen sets of $\mathbb{R}$ are $\mathbb{R}$ and $\varnothing$

Lemma 3 (Lemma 3.3.2) *Let S be a subset of $\mathbb{R}$ bounded above and let $p = \sup S$. If S is closed, then $p \in S$*

Lemma 4 *Let S be a subset of $\mathbb{R}$ bounded above and let $p = \sup S$. If S is open, then $p \notin S$*

Proof of Lemma 4

S is open and $p \in S$, we can choose an open interval $(a, b) \subseteq S$ containing p. Hence, $p < b$ and there exists q such that $p < q < b$. Contradiction to the assumption that p is the supremum of S

Main proof

Suppose A, B are non-empty clopen sets in $\mathbb{R}$ such that $A \cap B = \varnothing$ and $A \cup B = \mathbb{R}$. Choose $a_1 \in A$ and $b_1 \in B$, without loss of generality, assume that $a_1 < b_1$.

Consider the closed set $S = A \cap [a_1, b_1]$ and its supremum $p = \sup S$

S is closed and bounded above by b_1, by Lemma 3, $p \in S$. Furthermore, $b_1 \notin A$ implies $b_1 \notin S = A \cap [a_1, b_1]$, we have the strict inequality $p < b_1$

Now we will construct an element $t \in S$ that is greater than p then conclude the contradiction. $p \in S \subseteq A$, A is open then there exists an open interval $(a_2, a_3) \subseteq A$ such that $p \in (a_2, a_3) \subseteq A$. By the strict inequality $p < b_1$, we can choose $t \in (p, \min(a_3, b_1)) \subseteq (a_2, a_3) \subseteq A$. We also have $t \in (p, \min(a_3, b_1)) \subseteq [a_1, b_1]$. Hence, $p < t \in A \cap [a_1, b_1] = S$. Contradiction

Exercise 4.1.11

Let A, B be connected subspaces of a topological space $(X, \mathcal{T})$. If $A \cap B \neq \varnothing$, prove that the subspace $A \cup B$ is connected.

Main proof

We will prove the statement by contradiction, first we assume that $A \cap B \neq \varnothing$ and $A \cup B$ is disconnected. Hence, we can find $P, Q \in A \cup B$ such that $P \neq \varnothing$, $Q \neq \varnothing$, $P \cup Q = A \cup B$, and $P \cap Q = \varnothing$. P, Q are corresponding to two open sets in X, namely $P = O_P \cap (A \cup B)$ and $Q = O_Q \cap (A \cup B)$.

Consider 2 pairs of sets: $(O_P \cap A, O_Q \cap A)$ and $(O_P \cap B, O_Q \cap B)$. Claim that there must be at least a pair with no empty set.

(case 1) $(O_P \cap A) = (O_P \cap B) = \varnothing$

$(O_P \cap A) \cup (O_P \cap B) = \varnothing$ implies $P = O_P \cap (A \cup B) = \varnothing$, contradiction

(case 2) $(O_P \cap A) = (O_Q \cap B) = \varnothing$.

$O_P \cap A = \varnothing$ implies that

$$\begin{aligned} P &= O_P \cap (A \cup B) \\ &= O_P \cap (A \cup B \setminus A) \\ &= (O_P \cap A) \cup (O_P \cap B \setminus A) \\ &= O_P \cap B \setminus A \subseteq B \setminus A \end{aligned}$$

$O_Q \cap B = \varnothing$ implies that

$$\begin{aligned} Q &= O_Q \cap (A \cup B) \\ &= O_Q \cap (A \setminus B \cup B) \\ &= (O_Q \cap A \setminus B) \cup (O_Q \cap B) \\ &= O_Q \cap A \setminus B \subseteq A \setminus B \end{aligned}$$

Hence, an element $x \in A \cap B$ is not in either P or Q, contradiction

Therefore, there must be at least a pair in $(O_P \cap A, O_Q \cap A)$ and $(O_P \cap B, O_Q \cap B)$ is both non-empty sets. Without loss of generality, assume that $O_P \cap A, O_Q \cap A$ are both non-empty.

$$\begin{aligned} (O_P \cap A) \cup (O_A \cap A) &= (O_P \cup O_Q) \cap A \\ &= A && \text{since } A \cup B \subseteq O_P \cup O_Q \end{aligned}$$

$$\begin{aligned} (O_P \cap A) \cap (O_A \cap A) &= (O_P \cap O_Q) \cap A \\ &= \varnothing && \text{since } O_P \cap O_Q \text{ is outside of } A \cup B \end{aligned}$$

Exercise 4.1.15

The closed interval $[a, b]$ for $a, b \in \mathbb{R}$ is connected.

Let's recognize all the open sets and closed sets in $[a, b]$ first. O is an open set in $[a, b]$ if and only if $O = O_{\mathbb{R}} \cap [a, b]$ for an open set $O_{\mathbb{R}}$ in $\mathbb{R}$. C is a closed set in $[a, b]$ if and only if $C = [a, b] \setminus O_{\mathbb{R}}$ for an open set $O_{\mathbb{R}}$ in $\mathbb{R}$

Lemma 5 *If C is a closed set in $[a, b]$, C is also a closed set in $\mathbb{R}$.*

Proof of Lemma 5

$\mathbb{R} \setminus C = (\mathbb{R} \setminus [a, b]) \cup O_{\mathbb{R}}$, union of two open sets in $\mathbb{R}$

Main proof

Similar to exercise 4.1.11, let prove the statement by contradiction. Let A, B be clopen sets in $[a, b]$ such that $A \cap B = \varnothing$ and $A \cup B = [a, b]$. Choose $a_1 \in A$ and $a_2 \in B$, without loss of generality, assume the strict inequality $a < a_1 < b_1 < b$ since A, B cannot be singleton sets.

A is a closed set in $\mathbb{R}$, consider the closed set $S = A \cap [a_1, b_1]$ in $[a, b]$ and its supremum $p = \sup S$ in $\mathbb{R}$.

S is closed and bounded above by b_1 in $\mathbb{R}$, by Lemma 3, $p \in S$. Furthermore, $b_1 \notin A$ implies $b_1 \notin S = A \cap [a_1, b_1]$, we have the strict inequality $a < a_1 \leq p < b_1 \leq b$

Now we will construct an element $t \in S$ that is greater than p then conclude the contradiction. $p \in S \subseteq A = O_A \cap [a, b] = (O_A \cap (a, b)) \cup (O_A \cap \{a, b\})$ where O_A is an open set in $\mathbb{R}$. $a < p < b$ implies p in the open set $O_A \cap (a, b) \subseteq A$ in $\mathbb{R}$. By the strict inequality $p < b_1$, we can choose $t \in (p, \min(a_3, b_1)) \subseteq (a_2, a_3) \subseteq O_A \cap (a, b) \subseteq A$. We also have $t \in (p, \min(a_3, b_1)) \subseteq [a_1, b_1]$. Hence, $p < t \in A \cap [a_1, b_1] = S$. Contradiction

Comment: $O_A \cap (a, b)$ is the interior of $A = O_A \cap [a, b]$ in $\mathbb{R}$

Exercise 4.1.17.v

Let $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. Define a set $C \subseteq \mathbb{R}$ to be closed if $C = A \cup T$ where A is closed in $\mathbb{R}$ and $T \subseteq S$. The complements of these closed sets form a topology $\mathcal{T}$ on $\mathbb{R}$ which is Hausdorff but not regular.

Definition 1 (Hausdorff space or T_2-space) *A topological space $(X, \mathcal{T})$ is said to be Hausdorff (or T_2-space) if given any pair of distinct points a, b in X there exist open sets A, B such that $a \in A$, $b \in B$, and $A \cap B = \varnothing$*

Definition 2 (Regular space) *A topological space $(X, \mathcal{T})$ is said to be regular space if any closed set A and any point $x \in X \setminus A$, there exist open sets U, V such that $x \in U$, $A \subseteq V$, and $U \cap V = \varnothing$.*

Lemma 6 *For any index set J and $A_j \cap B_j = \varnothing$ for all $j \in J$*

$$\bigcap_{j \in J} A_j \cup B_j = \bigcup_{J_A \in \mathcal{P}(J)} \left[\left(\bigcap_{j \in J_A} A_j \right) \cap \left(\bigcap_{j \in J \setminus J_A} B_j \right) \right]$$

Proof of Lemma 6

Proof

For all $x \in X = \bigcap_{j \in J} A_j \cup B_j$, for each $j \in J$, x must be either in A_j or B_j. Let $J_A(x) = \{j : j \in J, x \in A_j\} \subseteq \mathcal{P}(J)$ be the set of indices where $x \in A_j$ and let $J_B(x) = J \setminus J_A(x)$. So that

$$x \in \left(\bigcap_{j \in J_A(x)} A_j \right) \cap \left(\bigcap_{j \in J \setminus J_A(x)} B_j \right)$$

On the other hand, $\bigcap_{j \in J_A} A_j \subseteq \bigcap_{j \in J_A} A_j \cup B_j$ and $\bigcap_{j \in J \setminus J_A} B_j \subseteq \bigcap_{j \in J \setminus J_A} A_J \cup B_j$, we have

$$\begin{aligned}
\left(\bigcap_{j \in J_A(x)} A_j \right) \cap \left(\bigcap_{j \in J \setminus J_A(x)} B_j \right) &\subseteq \left(\bigcap_{j \in J_A(y)} A_j \cup B_j \right) \cap \left(\bigcap_{j \in J \setminus J_A(y)} A_j \cup B_j \right) \\
&= \bigcap_{j \in J} A_j \cup B_j \\
&= X
\end{aligned}$$

Invoke the theorem 1

$\mathcal{T}$ is a topology

Take $A = \varnothing$ and $T = \varnothing$, then $C = A \cup T = \varnothing$. So $\mathbb{R}$ is an open set in $\mathcal{T}$. Take $A = \mathbb{R}$, then $C = A \cup T = \mathbb{R}$. So $\varnothing$ is an open set in $\mathcal{T}$

$$\begin{aligned}
\mathbb{R} \setminus C_1 \cap \mathbb{R} \setminus C_2 &= \mathbb{R} \setminus (A_1 \cup T_1) \cap \mathbb{R} \setminus (A_2 \cup T_2) \\
&= \mathbb{R} \setminus ((A_1 \cup T_1) \cup (A_2 \cup T_2)) \\
&= \mathbb{R} \setminus ((A_1 \cup A_2) \cup (T_1 \cup T_2))
\end{aligned}$$

$A_1 \cup A_2$ is an closed set in the euclidean topology, $T_1 \cup T_2 \subseteq S$. Hence the intersection of two open sets in $\mathcal{T}$ is an open set in $\mathcal{T}$

Let J be an index set, a union of open sets in $\mathcal{T}$ has the form

$$\bigcup_{j \in J} \mathbb{R} \setminus (A_j \cup T_j) = \mathbb{R} \setminus \bigcap_{j \in J} A_j \cup T_j$$

We need to prove that $C = \bigcap_{j \in J} A_j \cup T_j$ can be written in the form $A \cup T$ where A is a closed set in the euclidean topology and $T \subseteq S$. Invoke the lemma 6

$$\begin{aligned}
\bigcap_{j \in J} A_j \cup T_j &= \bigcap_{j \in J} A_j \cup T_j \setminus A_j \\
&= \bigcup_{J_A \in \mathcal{P}(J)} \left[\left(\bigcap_{j \in J_A} A_j \right) \cap \left(\bigcap_{j \in J \setminus J_A} T_j \setminus A_j \right) \right]
\end{aligned}$$

We split $\mathcal{P}(J)$ into two groups: (1) $\{J\} \in \mathcal{P}(J)$ and (2) $\mathcal{P}(J) \setminus \{J\}$.

$$\bigcap_{j\in J} A_j \cup T_j = \bigcap_{j\in J} A_j \cup \bigcup_{J_A \in \mathcal{P}(J)\setminus J} \left[\left(\bigcap_{j\in J_A} A_j \right) \cap \left(\bigcap_{j\in J\setminus J_A} T_j \setminus A_j \right) \right]$$

The set $\bigcap_{j\in J} A_j$ is a union of closed set in euclidean space hence a closed set in euclidean space.

We further have

$$\left(\bigcap_{j\in J_A} A_j \right) \cap \left(\bigcap_{j\in J\setminus J_A} T_j \setminus A_j \right) \subseteq \bigcap_{j\in J\setminus J_A} T_j \setminus A_j \subseteq S$$

for all $J_A \in \mathcal{P} \setminus J$ since $J \setminus J_A$ is non-empty. Take A and T as follows

$$A = \bigcap_{j\in J} A_j$$

$$T = \bigcup_{J_A \in \mathcal{P}(J)\setminus J} \left[\left(\bigcap_{j\in J_A} A_j \right) \cap \left(\bigcap_{j\in J\setminus J_A} T_j \setminus A_j \right) \right]$$

So, $C = A \cup T$ which is a union of a closed set in the euclidean topology and a subset of S

$\mathcal{T}$ is Hausdorff

Any closed set in the euclidean topology is closed in $\mathcal{T}$ by taking $T = \varnothing$ in the form of closed set in $\mathcal{T}$: $C = A \cup T$. Hence, any open set in the euclidean topology is open in $\mathcal{T}$.

Given any two point $a < b \in \mathbb{R}$, take $A = (-\infty, \frac{a+b}{2})$ and $B = (\frac{a+b}{2}, +\infty)$

$\mathcal{T}$ is not regular

We will construct an example where it is not able to construct U, V

Suppose that $\mathcal{T}$ is a regular space, let $A = S = \{\frac{1}{n} : n \in \mathbb{N}\}$ and $x = 0$. There must be an open set V that contains 0 but not S. All open sets in $\mathcal{T}$ has the form

$$O = \mathbb{R} \setminus (A \cup T) = \mathbb{R} \setminus A \cap \mathbb{R} \setminus T$$

where A is a closed set in the euclidean topology and $T \subseteq S$.

$0 \in \mathbb{R} \setminus A$ an open set in the euclidean topology, there must be an open interval (a, b) such that $a < 0 < b$, hence that interval will contain some element of S, i.e: $\{\frac{1}{n} : n \in \mathbb{N}, n > \frac{1}{b}\}$

$\mathcal{T}$ is a topology (shorten approach)

I came across a short reasoning for the last statement [1]

$C = \bigcap_{i \in I} A_i \cup T_i$ where each A_i is Euclidean-closed and each $T_i \subseteq S$. Let $A = \bigcap_{i \in I} A_i$; certainly A is Euclidean-closed.

If $x \in C \setminus A$, there exists A_i such that $x \notin A_i$. But $x \in C$, so $x \in A_i \cup T_i$. Therefore, $x \in T_i \subseteq S$

Next

[1] https://math.stackexchange.com/a/65558/700122